SYNTHETIC DATA CREATION

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PROBLEMS WITH DATA

- Not enough data
- Closed-source
- Unavailable
- Costly
- Time-consuming

DEFINITION

Synthetic data is the data generated using a purpose-built mathematical model or algorithm to solve a (set of) data science task(s). Keep in mind that synthetic data only mimics the real data and is not generated by real events. Ideally, the synthetic data should have the same statistical properties as the real data it is supplementing. It has many uses, such as improving AI models, protecting sensitive data, and mitigating bias.

REAL DATA PROBLEMS

- Messy and hard to deal with
- Privacy issues
- Can be biased
- Data collection and annotation can be expensive
- Confidentiality (medical data)

ADVANTAGES OF SYNTHETIC DATA

- Synthetic data are generated in a structured form, and hence, they are easy to deal with.
- Companies can train synthetic data generation models that learn the distribution of the original data but don't reveal anything about individual data points in the original data and hence maintain privacy. A similar approach can be taken for medical data.
- We can train the data generator model to generate de-biased data.
- Synthetic data can be augmented with real data to make the models or applications more robust.

HOW?

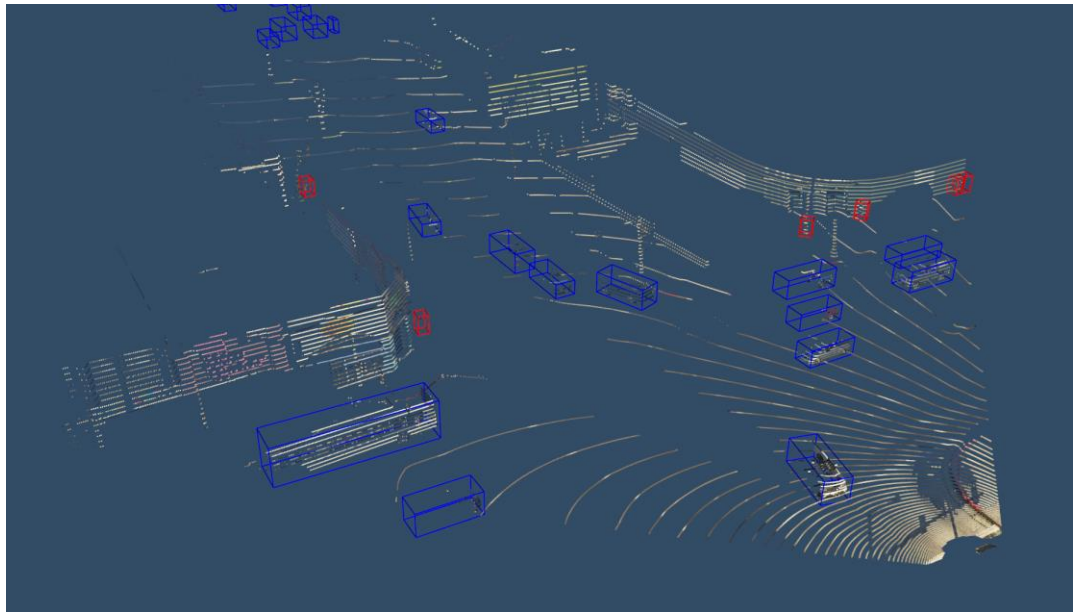
- **CAD & Blender**: Allows the creation of photorealistic image datasets of 3D scenes while controlling parameters. It enables computing metrics by comparing the synthesized data to the ground truth (generation parameters). It is a very robust method but limited in generation quality, diversity, and quantity.
- **Deep generative models (Transformers/GANs/Diffusion models):** Allow expanding a dataset, tackling data imbalance, and solving privacy issues. Very convenient and powerful but can create datasets with biases, incoherence, and repetitiveness, which induces an important overtraining risk and produces a restricted set of predictions.

SYNTHETIC DATASETS

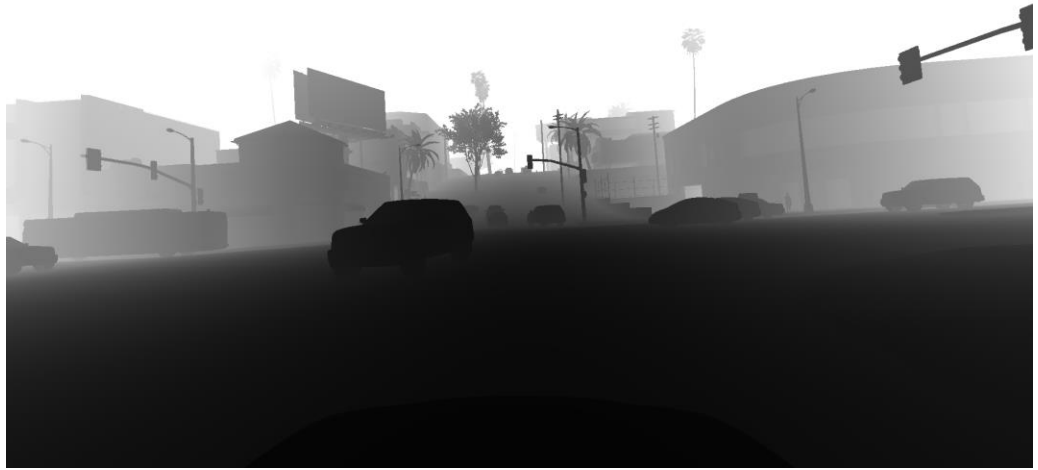
- Low level problems
  - optical flow and motion analysis
    - MIddleBury
    - MPI-Sintel
- High level
  - semantic segmentation for autonomous driving
    - Virtual KITTI 2
    - CARLA
    - GTA 5
  - Indoor simulation and Navigation
    - Habitat
    - Minos
  - Face Recognition
    - FFHQ

SOME EXAMPLES OF GTA 5 SYNTHETIC DATASET

Image



Lidar point cloud



Linearized depth map

Object instance segmentation

Notebook DCGAN